Dr. Sagar H P

Data Scientist | Research Engineer

About me

I hold a Ph.D., from the **Indian Institute of Technology – Bombay.** My thesis is focused on the integration of **Triboelectric devices for Industry 4.0** applications. My research led to the development of self-powering triboelectric vibration sensors for condition monitoring. I have published **8 peer-reviewed journal articles** and **6 conference papers** across multiple fields.

Currently, as a **Data Scientist** at Halliburton, I leverage AI/ML to craft solutions for diverse organizational needs. I'm now pursuing a challenging role to apply my multidisciplinary background in solving complex real-world multiphysics problems to drive innovations and create an impactful success story.

Contact

Date Of Birth: 3rd July 1989

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linkedin.com/in/hpsagar

github.com/hprutvisagar (code repo)

hprutvisagar.github.io (e-portfolio)

#506, Siri Dew Drops Apartments
7th main Nandakumar layout,
Bengaluru - 560061, India

EDUCATION

Doctorate of Philosophy **2016-2022**
Indian Institute of Technology Bombay
Thesis: Triboelectric effect driven self-powered vibration sensors and wind energy harvesting device for enabling industry 4.0.
TATA fellowship awardee
Grade: 8.59 CGPA

M.Tech - Nanotechnology **2013-2015**
National Institute of Technology Karnataka, Surathkal
Thesis: Design and analysis of 1D silicon photonic crystal-based strain and mass sensor.
Collaborative research with IISc Bengaluru
Grade: 9.5 CGPA

B.E. - Mechanical Engineering **2007-2011**
Visvesvaraya Technological University
(Dr. Ambedkar Institute of Technology)
Thesis: Design and development of automated bike washer unit.
Grade: 71.8%

WORK EXPERIENCE (6+ years)

Data Scientist **June, 2021-Aug, 2023**
Halliburton

- Managed a comprehensive proof-of-concept initiative with complete ownership of backend model development, high-level system designing, and frontend optimizations. I built a resilient regression-based model which can estimate the cloud-application runtime with >95% confidence.
- Developed and implemented a variety of machine learning models including regression, classification, natural language processing (NLP), anomaly detection, and failure prediction.
- I contribute in company outreach by preparing technical abstracts for conferences to showcase Halliburton's innovations in oil and gas industry.
- Translated intricate business requirements into well-defined data science problem statements by collaborating closely with cross-functional teams to ensure alignment between technical solutions and strategic business objectives.

Senior Structural Analyst **June, 2019 - June, 2021**
Gorgonian Technologies

- Here, I was actively involved in research and development of small-scale wind turbines. My major responsibilities were to perform CFD analysis and check the feasibility of the designer's ideas.

Mechanical engineer **Feb, 2019 - Feb, 2020**
Aumeesh Technologies

- Here, I worked on a KAFO (leg prosthetic) product to optimize its mechanical systems based on GAIT analysis.

Mechanical (CAD) Engineer **Sept, 2011 - Aug, 2013**
HCL technologies

- Provided CAD support for semiconductor domain clients. I specialized in cable routing path optimization, and creating its flat-board drawings for manufacturing.

Soft Skills and Strengths

Creativity | Curiosity | Flexibility | Self Confidence | Ability to Plan and Organize | Autonomy | Adaptability | Eye for Details | Problem Solving | Team Working | Love Learning New Things | Leadership | Good Communication | Managing Information | Diplomacy | Good Listener | Patience

Technologies

Tribo-electric devices | Vibration sensors | Savonius wind turbines | Condition monitoring | Electromechanical devices | Screen printing | Data science | Machine Learning | Artificial Intelligence

Languages

Kannada - Native

English - Professional Fluent

Hindi - Conversational

Telugu - Conversational

SELECTED PUBLICATIONS

Journal Article
Total 8

Self powering vibration sensor based on a cantilever system with a single electrode mode triboelectric nanogenerator, **Sagar Hosangadi Prutvi**, Mallikarjuna Korrapati, and Dipti Gupta, *Measurement Science and Technology 33 (7), 075115 (2022)*, 10.1088/1361-6501/ac5b2b

Triboelectric effect based self-powered compact vibration sensor for predictive maintenance of industrial machineries, Hosangadi Prutvi Sagar, Sunil Meti, Udaya K Bhat, and Dipti Gupta*, *Measurement Science and Technology 32 (9), 095119 (2021)*, 10.1088/1361-6501/abe6d2

Transient dynamic distributed strain sensing using photonic crystal waveguides, **Hosangadi Prutvi Sagar**, Vignesh Mahalingam, Debiprosad Roy Mahapatra*, Gopalkrishna Hegde, Sathyanarayana Hanagud, and Mohammad Rizwanur Rahman, *Applied Optics 56 (28), 7877-7885 (2017)*, 10.1364/AO.56.007877

Conference & symposiums
Total 6

Graphene Integrated Waveguide for Molecular Sensing, **Sagar H P**, and MR Rahman,, *International Engineering Symposium (IES) - 2015,* ***Kumamoto University, Kumamoto, Japan***

SKILLS

Research & Development	Matlab | Rapid Prototyping | SEM & AFM | Thermal Evaporator | Plasma ashers | Screen printing | wet bench experimentations | Instrumentations | Analog circuits | Arduino
Data Science	Python | AWS (Sagemaker and Canvas) | Orange | TensorFlow | Keras | Pandas | Seaborn | Numpy
Simulation & Analysis	COMSOL Multiphysics | Mathematica | Ansys | Ansa | LS-Dyna | LT-spice | Hypermesh | LAMMPS
3D modeling	Creo | ProE | Autocad | Solidworks | Catia

CERTIFICATES

- NLP - Natural Language Processing with Python (*2022*)
- Python for Time Series Data Analysis (*2022*)
- Python for Data Science and Machine Learning bootcamp (*2021*)

- Generative AI fundamentals (*2023*)